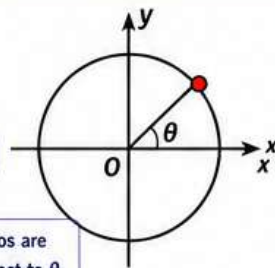


MATHS PATHSHALA

CLASS 10 – TRIGONOMETRY FORMULA SHEET

—♦ Your Success, Our Mission ♦—

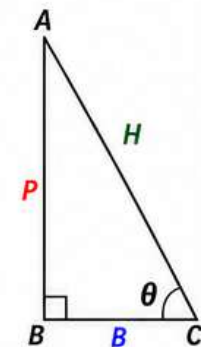
All ratios are with respect to θ .



1. TRIGONOMETRIC RATIOS

In $\triangle ABC$, right-angled at B

- $\sin \theta = \frac{P}{H}$
- $\cos \theta = \frac{B}{H}$
- $\tan \theta = \frac{P}{B}$
- $\operatorname{cosec} \theta = \frac{H}{P}$
- $\sec \theta = \frac{H}{B}$
- $\cot \theta = \frac{B}{P}$



P = Perpendicular
B = Base
H = Hypotenuse

2. RECIPROCAL RELATIONS

$$\sin \theta \cdot \operatorname{cosec} \theta = 1$$

$$\cos \theta \cdot \sec \theta = 1$$

$$\tan \theta \cdot \cot \theta = 1$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

3. QUOTIENT RELATIONS

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

4. PyTHAGORAS IDENTITY (IMPORTANT)

$$\star \sin^2 \theta + \cos^2 \theta = 1$$

$$\star 1 + \tan^2 \theta = \sec^2 \theta$$

$$\star 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$



Remember

✓ Change the function, value becomes 1.

✓ Very helpful in solving questions.

5. STANDARD VALUES TABLE

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

6. SIGNS OF TRIGONOMETRIC RATIOS (ALL RATIOS)

Second Quadrant (90° to 180°) $\sin +$ $\cos -$ $\tan -$	First Quadrant (0° to 90°) $\sin +$ $\cos +$ $\tan +$
Third Quadrant (180° to 270°) $\sin -$ $\cos -$ $\tan +$	Fourth Quadrant (270° to 360°) $\sin -$ $\cos +$ $\tan -$

7. ALLIED ANGLE IDENTITIES

$\sin (90^\circ - \theta) = \cos \theta$	$\sin (180^\circ - \theta) = \sin \theta$
$\cos (90^\circ - \theta) = \sin \theta$	$\cos (180^\circ - \theta) = -\cos \theta$
$\tan (90^\circ - \theta) = \cot \theta$	$\tan (180^\circ - \theta) = -\tan \theta$
$\cot (90^\circ - \theta) = \tan \theta$	$\cot (180^\circ - \theta) = -\cot \theta$
$\sec (90^\circ - \theta) = \operatorname{cosec} \theta$	$\sec (180^\circ - \theta) = -\sec \theta$
$\operatorname{cosec} (90^\circ - \theta) = \sec \theta$	$\operatorname{cosec} (180^\circ - \theta) = \operatorname{cosec} \theta$

8. OTHER IMPORTANT IDENTITIES

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$(\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta - \cot \theta) = 1$$

EXAM TIPS

- ✓ Learn standard values thoroughly.
- ✓ Draw a neat diagram.
- ✓ Use identities smartly.
- ✓ Practice previous year questions.
- ✓ Identify the quadrant before applying signs.

“Practice is the key to Mastery.
Keep Learning!”



TRIGONOMETRY SHORTCUT

To remember ratios:
 $\sin = P/H$, $\cos = B/H$, $\tan = P/B$
Then use reciprocal & identities for all other ratios.

